# LLMOps Assignment 2: API Cost, Latency and Prompt Version Management

EduAssist AI academic Q&A assistant

## *1. Introduction*

MLOps manages datasets, training, deployment and monitoring of models. LLMOps extends this to LLM applications, where prompts, model versions and output guardrails are versioned artifacts and every request has a token-based cost and a latency. This project measures both for EduAssist AI's Q&A assistant, then promotes the best of three prompt versions in MLflow.

## *2. Setup and deviation from the brief*

OpenAI API access was unavailable (calls failed with an insufficient_quota error), so, as section 7 of the brief allows, the project uses a locally hosted open-source model: llama3.1:8b, served by Ollama on Google Colab (Tesla T4) and called through its OpenAI-compatible chat completions endpoint with the openai SDK, so no API key is needed. Temperature was 0; token counts come from each response's usage field. Local inference has no bill, so cost is estimated from published hosted pricing for the same model family: DeepInfra (deepinfra/fp8) via OpenRouter, meta-llama/llama-3.1-8b-instruct, listed price, https://openrouter.ai/meta-llama/llama-3.1-8b-instruct, checked on 2026-11-04, at 0.02 USD per million input tokens and 0.04 USD per million output tokens.

## *3. Task 1: latency*

Prompt v2, five questions. Latency in seconds: mean 10.52, median 10.63, minimum 7.72, maximum 13.26, standard deviation 2.45. The slowest request was Q5 at 13.26 s with 426 output tokens, while the other four averaged 9.84 s and 342 output tokens. Latency and output tokens are strongly related across the five requests (r = 0.98); the model produced about 34 tokens per second, and tokens are generated one after another, so longer answers take longer. There is no internet round trip (Tesla T4, local server); other possible reasons are prompt length (49 to 56 input tokens) and GPU scheduling. Five samples are only indicative.

## *4. Task 2: tokens and cost*

Average cost was 0.00001540 USD per request; projected monthly cost for 10,000 requests is 0.15 USD. Shorten responses. Output tokens made up 93% of the measured cost because they are priced 2.0 times higher than input tokens. Asking for concise answers and setting max_tokens to 200 would cut average output from 359 to about 200 tokens, lowering the average cost per request from 0.00001540 to 0.00000905 USD (41% less, about 0.0634 USD per 10,000 requests). Shorter output also reduces latency; check answer quality first.

## *5. Task 3: prompt versions in MLflow*

Experiment eduassist_llmops_assignment holds one run per version; prompts are registered as eduassist_qa_prompt versions 1 to 3 with commit messages.

| Version | Avg latency (s) | Avg input tok | Avg output tok | Avg cost (USD) | Valid JSON | Recommended? |
|---|---|---|---|---|---|---|
| v1 | 16.91 | 30.6 | 500.4 | 0.00002063 | 0% | No |
| v2 | 13.42 | 52.6 | 358.6 | 0.00001540 | 0% | No |
| v3 | 4.48 | 86.6 | 115.2 | 0.00000634 | 100% | Yes |

## *6. Promotion justification*

Prompt v3 (registry version 3) was promoted to the production alias because the promotion rule ranks the valid_json rate first and average cost second. Valid JSON rates were 0% for v1, 0% for v2 and 100% for v3. This matters because downstream code must parse every response.

Average latency was 16.91 s for v1, 13.42 s for v2 and 4.48 s for v3. Average input tokens were 31, 53 and 87, and average output tokens were 500, 359 and 115. Average cost per request was 0.00002063, 0.00001540

and 0.00000634 USD. Version v3 generated the fewest output tokens and was also the fastest and cheapest, because output tokens are priced 2.0 times higher than input tokens and generation time grows with output length.

On quality, v1 and v2 answers averaged 380 and 289 words of free text, while the v3 answer field averaged 44 words plus 3.0 key points and a confidence label (high 5). v3 is shorter but adds structure that v1 and v2 lack. These are measurable proxies, not a grade of correctness.

The JSON guarantee did not cost extra here: v3 added 34 input tokens per request compared with v2, but its shorter outputs made it cheaper, so the structured format came with lower cost and latency.

A valid JSON rate measured on five samples does not prove reliability at scale, so validity should be monitored, keeping v1 and v2 available for rollback through aliases.